Fundamental solution sets and the Wronskian

We just looked at how to use the method of undetermined coefficients and the method of variation of parameters to find the solution to a second order linear nonhomogeneous equation.

When we used variation of parameters, we took a fundamental set of solutions $\{y_1, y_2\}$ from the complementary solution,

Complementary solution $y_c(x)$	**Fundamental solution set** $\{y_1, y_2\}$
$y_c(x) = c_1 e^{r_1 x} + c_2 e^{r_2 x}$	$\{y_1, y_2\} = \{e^{r_1 x}, e^{r_2 x}\}$
$y_c(x) = c_1 e^{r_1 x} + c_2 x e^{r_1 x}$	$\{y_1, y_2\} = \{e^{r_1 x}, x e^{r_1 x}\}$
$y(x) = e^{\alpha x}(c_1 \cos(\beta x) + c_2 \sin(\beta x))$	$\{y_1, y_2\} = \{e^{\alpha x} \cos(\beta x), e^{\alpha x} \sin(\beta x)\}$

and then we plugged that solution set into a system of linear equations.

$$u_1' y_1 + u_2' y_2 = 0$$

$$u_1' y_1' + u_2' y_2' = g(x)$$

We solved this system for u_1' and u_2', then integrated those values to find u_1 and u_2, which we then used to build a particular solution for the differential equation.

At this point though, we want to expand on the idea of a fundamental set of solutions, and then introduce a different way to look at the method of variation of parameters.

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Fundamental set of solutions

Technically, we say that a set of solutions to a differential equation is a **fundamental set of solutions** when the Wronskian of those solutions is non-zero.

The **Wronskian** W of a solution set $\{y_1, y_2\}$ is the determinant of the set of solutions and their derivatives.

$$W(y_1, y_2)(t_0) = \begin{vmatrix} y_1(t_0) & y_2(t_0) \\ y_1'(t_0) & y_2'(t_0) \end{vmatrix} = y_1(t_0)y_2'(t_0) - y_2(t_0)y_1'(t_0)$$

So if $y_1(t)$ and $y_2(t)$ are solutions to the differential equation, and if the Wronskian is non-zero $W \neq 0$, then we say that the solutions $y_1(t)$ and $y_2(t)$ are **linearly independent**, which means that they're not constant multiples of one another. When the Wronskian is non-zero and the solution set is therefore linearly independent, we call $\{y_1, y_2\}$ a fundamental set of solutions. If we have a fundamental set of solutions, we can conclude that the general solution is

$$y(t) = c_1 y_1(t) + c_2 y_2(t)$$

In other words, we've actually been assuming up to this point that the solution set for a second order equation is a fundamental set, because we've been using any solution set we found to build the general solution, without confirming that $W \neq 0$.

So let's circle back to take a look at the case of distinct real roots, and confirm now that $\{y_1, y_2\} = \{e^{r_1 x}, e^{r_2 x}\}$ is a fundamental set of solutions, by

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plugging the solutions and their derivatives into the Wronskian determinant.

$$W(e^{r_1x}, e^{r_2x})(x) = \begin{vmatrix} e^{r_1x} & e^{r_2x} \\ r_1e^{r_1x} & r_2e^{r_2x} \end{vmatrix}$$

$$W(e^{r_1x}, e^{r_2x})(x) = (e^{r_1x})(r_2e^{r_2x}) - (e^{r_2x})(r_1e^{r_1x})$$

$$W(e^{r_1x}, e^{r_2x})(x) = r_2e^{r_1x+r_2x} - r_1e^{r_1x+r_2x}$$

$$W(e^{r_1x}, e^{r_2x})(x) = e^{(r_1+r_2)x}(r_2 - r_1)$$

We need to confirm that this Wronskian value is non-zero. The only way $W = 0$ is if $e^{(r_1+r_2)x} = 0$ and/or $r_2 - r_1 = 0$. The exponential function is never zero. And remember that this is the case with distinct real roots, which means $r_2 \neq r_1$, so $r_2 - r_1 \neq 0$.

In other words, the Wronskian of the solution set $\{y_1, y_2\} = \{e^{r_1x}, e^{r_2x}\}$, assuming $r_1 \neq r_2$, will be a fundamental set of solutions to the second order equation.

And similar logic applies to the solution set for equal real roots and for complex conjugate roots, which is why we've been able to rely on $\{y_1, y_2\} = \{e^{r_1x}, xe^{r_1x}\}$ to form the general solution to a second order equation with equal real roots, and rely on $\{y_1, y_2\} = \{e^{\alpha x}\cos(\beta x), e^{\alpha x}\sin(\beta x)\}$ to form a general solution to a second order equation with complex conjugate roots.

The reliable set

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There's a set of initial conditions we can use that will always give us a fundamental set of solutions for $y'' + p(t)y' + q(t)y = 0$, which is $y(t_0) = 1$ and $y'(t_0) = 0$, and $y(t_0) = 0$ and $y'(t_0) = 1$.

The reason these two pairs of initial conditions produce a fundamental set of solutions is because the Wronskian they make will never be 0.

$$W(y_1, y_2)(t_0) = \begin{vmatrix} y_1(t_0) & y_2(t_0) \\ y_1'(t_0) & y_2'(t_0) \end{vmatrix} = \begin{vmatrix} 1 & 0 \\ 0 & 1 \end{vmatrix} = 1 - 0 = 1 \neq 0$$

Let's do an example with this set of initial conditions that we know will reliably produce a fundamental set of solutions.

Example

Use the reliable set of initial conditions to find a fundamental set of solutions for the second order differential equation.

$$y'' + 5y' + 6y = 0$$

Normally, we would find the general solution to this second order equation by solving the associated characteristic equation,

$$r^2 + 5r + 6 = 0$$

$$(r + 3)(r + 2) = 0$$

$$r = -3, \ -2$$

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and then plugging these values into the formula for the complementary solution with distinct real roots.

$$y_c(x) = c_1 e^{-3x} + c_2 e^{-2x}$$

The derivative of this complementary solution is

$$y_c'(x) = -3c_1 e^{-3x} - 2c_2 e^{-2x}$$

If we now plug in the initial conditions $y(x_0) = 1$ and $y'(x_0) = 0$, and $y(x_0) = 0$ and $y'(x_0) = 1$, we'll be able to generate another fundamental set of solutions. We'll use $x_0 = 0$, since we can choose any value of x_0 that we like, and using $x_0 = 0$ will be easiest.

For $y(0) = 1$, $\qquad 1 = c_1 + c_2$

For $y'(0) = 0$, $\qquad 0 = -3c_1 - 2c_2$

Then this system gives $c_1 = -2$ and $c_2 = 3$, so we could say that one solution in our fundamental set of solutions is

$$y_1(x) = -2e^{-3x} + 3e^{-2x}$$

The second set of initial conditions gives

For $y(0) = 0$, $\qquad 0 = c_1 + c_2$

For $y'(0) = 1$, $\qquad 1 = -3c_1 - 2c_2$

Then this system gives $c_1 = -1$ and $c_2 = 1$, so we could say that the other solution in our fundamental set is

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$$y_2(x) = -e^{-3x} + e^{-2x}$$

So one fundamental set of solutions is $\{y_1, y_2\} = \{e^{-3x}, e^{-2x}\}$, but another is

$$\{y_1, y_2\} = \{-2e^{-3x} + 3e^{-2x}, -e^{-3x} + e^{-2x}\}$$

and the complementary solution this set creates would be

$$y_c(x) = c_1(-2e^{-3x} + 3e^{-2x}) + c_2(-e^{-3x} + e^{-2x})$$

While this solution looks a lot messier than the solution we originally generated, and it *is* messier, the advantage of a solution generated this way is that it's a reliable fundamental set of solutions.

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